

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

CM2 - Economic Modelling (Paper A)

May 2025

INDICATIVE SOLUTION

- Q.1)** Answer: 4 Excess of Loss Reinsurance [2]
- Q.2)** Answer: 2 [2]
- Q.3)** Answer: 1 [2]
 $0.2 \cdot 100 + 0.7 \cdot 0 + 0.1 \cdot (-200) = 0$
- Q.4)** Answer: 2. When healthy individuals opt out of insurance, leaving a pool of high-risk individuals [2]
- Q.5)** Answer: 4. Investors can only invest in individual assets, not in portfolios [2]
- Q.6)** Answer: 3 Reduce the portfolio's overall risk without sacrificing expected return [2]
- Q.7)** Answer: 2. Allocate their income in such a way that the marginal utility per dollar spent is equal across all goods. [2]
- Q.8)** Answer: 1. The market portfolio has a higher expected return than the risk-free security. [2]
- Q.9)** Answer: 4. The inflation rate. [2]
- Q.10)** Answer: 3. The difference between the underlying asset price and the strike price. [2]
- Q.11)** Answer: 2. The drift of the process changes but the diffusion coefficient remains the same. [2]
- Q.12)** Answer: 3. Claims arrive according to a Poisson process, and claim sizes are i.i.d. random variables. [2]
- Q.13)** Answer: 4 A person becomes more satisfied with their income as they earn higher wages. [2]
- Q.14)** Answer: 5 $E(S) = 15.5$; $\text{Var}(S) = 139$ [2]

Let S_1 & S_2 be the total amount claimed on Type 1 & Type 2 policies respectively in one year. The total amount claimed on the whole portfolio is $S = S_1 + S_2$. Since all policies are assumed to be independent, the number of claims and the claim sizes for Type 1 policies are independent of those for Type 2 policies. Therefore, S_1 and S_2 are independent random variables.

- Mean of S : $E[S] = E[S_1 + S_2] = E[S_1] + E[S_2] = 7.5 + 8 = 15.5$
- Variance of S : Since S_1 and S_2 are independent, $\text{Var}(S) = \text{Var}(S_1 + S_2) = \text{Var}(S_1) + \text{Var}(S_2) = 75 + 64 = 139$

- Q.15)** Answer: 1 E [2]
- Q.16)** Answer: 3 [3]

Find the value of μ and σ

$$\exp(\mu + \sigma^2/2) = 9.070$$

$$\exp 2 (\mu + \sigma^2/2) \times (\exp \sigma^2 - 1) = 10.132^2$$

Solving for μ and σ

$$\mu = 1.80, \text{ and } \sigma = 0.90$$

$$P(X > 25) = 1 - \Phi \{(\log 25 - 1.80)/0.90\} = 0.0574$$

$$P(X > 30) = 1 - \Phi \{(\log 30 - 1.80)/0.90\} = 0.0376$$

Q.17) $E(X) = 4\%$; $\text{Var}(X) = 0.052\%$

Answer: 2

[3]

$$E[X] = (0 \times 0.10) + (2 \times 0.20) + (4 \times 0.40) + (6 \times 0.25) + (10 \times 0.05) = 4\%$$

$$\text{Var}[X] = 0.1 \cdot (4\% - 0\%)^2 + 0.2 \cdot (4\% - 2\%)^2 + 0.4 \cdot (4\% - 4\%)^2 + 0.25 \cdot (4\% - 6\%)^2 + 0.05 \cdot (4\% - 10\%)^2 = 0.052\%$$

Q.18) Answer 2. (a) 0.024%; (b) 70%; (c) 1.5%

[3]

Answer: (a) $\text{Semi var}(X) = 0.1 \cdot (4\% - 0\%)^2 + 0.2 \cdot (4\% - 2\%)^2 + 0.4 \cdot (4\% - 4\%)^2 = 0.024\%$

(b) Shortfall probability = $10\% + 20\% + 40\% = 70\%$

(c) Expected shortfall = $0.1 \cdot (5\% - 0\%) + 0.2 \cdot (5\% - 2\%) + 0.4 \cdot (5\% - 4\%) = 1.5\%$

Q.19) Answer: 1 Individual A is more likely to accept the gamble.

[3]

Individual A: $U_A(w) = w + 0.01w^2$

- Utility at current wealth (₹300): $U_A(300) = 300 + 0.01 \cdot (300)^2 = 300 + 0.01 \cdot (90000) = 300 + 900 = 1200$
- Expected utility with the gamble: $E[U_A(\text{gamble})] = 0.5 \cdot U_A(600) + 0.5 \cdot U_A(150)$
- $U_A(600) = 600 + 0.01 \cdot (600)^2 = 600 + 0.01 \cdot (360000) = 600 + 3600 = 4200$
- $U_A(150) = 150 + 0.01 \cdot (150)^2 = 150 + 0.01 \cdot (22500) = 150 + 225 = 375$
- $E[U_A(\text{gamble})] = 0.5 \cdot 4200 + 0.5 \cdot 375 = 2100 + 187.5 = 2287.5$
- Comparison: The expected utility with the gamble (2287.5) is greater than the utility at the current wealth (1200). Therefore, Individual A is likely to accept the gamble based on expected utility theory.

Individual B: $U_B(w) = \ln(w)$

- Utility at current wealth (₹300): $U_B(300) = \ln(300) \approx 5.7038$
- Expected utility with the gamble: $E[U_B(\text{gamble})] = 0.5 \cdot U_B(600) + 0.5 \cdot U_B(150)$
- $U_B(600) = \ln(600) \approx 6.3969$ $U_B(150) = \ln(150) \approx 5.0106$

Now, calculate the expected utility: $E[U_B(\text{gamble})] = 0.5 \cdot 6.3969 + 0.5 \cdot 5.0106 = 3.19845 + 2.5053 = 5.70375$

- Comparison: The expected utility with the gamble (5.70375) is very slightly less than the utility at the current wealth (5.7038). Therefore, Individual B is likely to reject the gamble based on expected utility theory (or be very close to indifferent).

Q.20) Answer: 2

[3]

$$P(1.05X < 100) = P(X < 100/1.05) = 1 - \{ \lambda / (\lambda + 100/1.05) \}$$

Putting $\lambda = 1000$ in above equation = 0.2727

Q.21) Answer: 3 ₹50,000

[3]

Expected cost to the insurer

- With maintenance = $0.01 \cdot 1,000,000 = 10,000$
- Without maintenance = $0.05 \cdot 1,000,000 = 50,000$

If both pay the same premium P, the firm compares

- With maintenance = $P + 10,000$
- Without maintenance = P

To incentivise maintenance

$P+10,000 < P + \text{expected loss of insurer saving} \Rightarrow 10,000 < 50,000 - 10,000 = 40,000 \Rightarrow$ Premium must reflect the savings to justify maintenance so max fair premium $= P \leq 50,000$

Q.22) Answer: 2 307.44

[3]

By put-call parity we know that

$$C_t + ke^{-r(T-t)} = P_t + S_t$$

$$P_t = 187.06 + 5,250 e^{-0.05 \times 0.5} - 5000 = 307.44$$

Q.23) Answer: 4 ₹500

[3]

- With precaution:
Probability of loss = 5% $\rightarrow$ Expected payout = $0.05 \times ₹10,000 = ₹500$
- Without precaution (due to moral hazard):
Probability of loss = 10% $\rightarrow$ Expected payout = $0.10 \times ₹10,000 = ₹1,000$
- Increase in expected payout:
 $₹1,000 - ₹500 = ₹500$

Q.24) Answer: 4 8.3%

[3]

$$\text{Portfolio Beta} = 0.4 \times 1.5 + 0.6 \times 0.75 = 1.05$$

$$\text{Apply CAPM} = R_p = R_f + \text{Beta} (R_m - R_f) = 2\% + 1.05 \times 6\% = 8.3\%$$

Q.25) $\text{Sigma}^2 = 1.178655$; $\mu = 6.095284$

Answer: 4

[3]

$$\exp(\mu + \sigma^2/2) = 800$$

$$\exp 2 (\mu + \sigma^2/2) \times (\exp \sigma^2 - 1) = 1200^2$$

Solving for μ and σ^2

$$\mu = 6.095284, \text{ and } \sigma = 1.178655$$

Q.26) Answer: 3 500, 545.6, 604.0, 653.6

[4]

414	460	482	488	500
453	506	526		
496	558			
540				
	1524	1008	488	500
1363	966	482		

Chain-ladder development factors:

	1.11812	1.04348	1.01245	1.02459
Forecasts:				
2000				500
2001			532.5	545.6
2002		582.3	589.5	604.0
2003	603.8	630.0	637.9	653.6

Q.27) Answer: 1 9.800, 13.736, 14.758, 15.312

[4]

Average cost per claim:

2000	8.254	9.328	9.504	9.660	9.800
2001	10.627	13.613	13.321		
2002	11.782	14.337			
2003	12.322				

Grossing-up method:

84.22%	95.19%	96.98%	98.57%	
77.37%	99.10%			13.736
79.83%				14.758
				15.312

Average: 80.47% Average: 97.14%

Q.28) Answer: 2 ₹12,104,500

	No of claims	Cost per claim	Projected loss
2000	500	9.8	4900
2001	545.6	13.736	7494.9
2002	604	14.758	8914.1
2003	653.6	15.312	10007.5
			31316.5

Claims paid to date: 19,212

Reserve= 12,104.6

i.e. ₹12,104,500

Q.29) A:15%,B:15%,C:15%; A:.02%,B:.375%,C:.0833%

Answer 5

[4]

- i) $E[RA] = (0.10 \times 0.4) + (0.15 \times 0.2) + (0.20 \times 0.4) = 15\%$
 $E[RB] = (0.10 \times 0.25) + (0.15 \times 0.70) + (0.40 \times 0.05) = 15\%$
 $E[RC] = (0.10 + 0.20) / 2 = 15\%$
- ii) $\text{Var}[RA] = (0.0025 \times 0.4) + (0 \times 0.2) + (0.0025 \times 0.4) = 20\%\%$
 $\text{Var}[RB] = (0.0025 \times 0.25) + (0 \times 0.70) + (0.0625 \times 0.05) = 37.5\%\%$
 $\text{Var}[RC] = (b-a)^2 / 12 = (0.20 - 0.10)^2 / 12 = \approx 8.33\%\%$

Q.30) Answer: 1 A:.10%,B:.0625%,C:.0417%; A:0.4,B:0.25,C:0.5

[4]

(i)

Semi-variance

Investment A $0.4 \times 0.05^2 = 0.001$ or 10%%

Investment B $0.25 \times 0.05^2 = 0.000625$ or 6.25%%

$$\begin{aligned}
 \text{Investment C} \quad & \int_{0.1}^{0.15} (0.15 - x)^2 10 \, dx \\
 &= 10 \left[0.15^2 x + \frac{x^3}{3} - 0.15x^2 \right]_{0.1}^{0.15} \\
 &= 10 \times (0.0011250 - 0.0010833) \\
 &= 0.000417 \text{ or } 4.17\%
 \end{aligned}$$

- (ii) Shortfall probability below 15%
 Investment A: 0.4, Investment B: 0.25, Investment C: 0.50

Q.31) Answer 1: 160.94

[4]

Solving for

$$\int_0^M 0.01 e^{-0.01x} dx = 0.8$$

$$M = \log 0.2 / (-0.01) = 160.94$$

Q.32) Answer: 1

[4]

The correlation coefficient ρ_{AB} is defined as:

$$\rho_{AB} = \frac{\text{Cov}(A, B)}{\sigma_A \sigma_B}$$

where:

$$\text{Cov}(A, B) = E[AB] - E[A]E[B]$$

Using the data given:

$$E[A] = 6.0\% \quad E[B] = 6.1\% \quad E[AB] = 19\%$$

$$\text{So: } \text{Cov}(A, B) = 19 - 6.0 \times 6.1 = -17.6\%$$

Also, using the data given:

$$\sigma_A = 2\%$$

$$\sigma_B = 9.0714\%$$

$$\text{So: } \rho_{AB} = \frac{-17.6}{2 \times 9.0714} = -0.970$$

Q.33) Answer: 4 Mean = 112.5; Variance = 12,694

[4]

Let X be the loss amount of a randomly chosen claim.

$$\text{Mean} = 0.25 \cdot \lambda_1 + 0.75 \cdot \lambda_2 = 0.25 \cdot 120 + 0.75 \cdot 110 = 30 + 82.5 = 112.5$$

$$\text{Variance} = 0.25 \cdot (\lambda_1^2 + \lambda_2^2) + 0.75 \cdot (\lambda_1^2 + \lambda_2^2) - 112.5^2 = 12694$$

Q.34) Answer: 3 36.59

[4]

We must find D such that

$$\int_0^D f(x) dx = 0.25$$

$$\text{where } f(x) = (\alpha \lambda^{-\alpha}) / (\lambda + x)^{\alpha+1}, \lambda = 300, \alpha = 2.5$$

Replacing the value and solving for D, we get D=36.59

Q.35) Answer: 3 ₹2.76

[4]

Given:

- $S = 100$
- $K = 105$
- $r = 0.05$
- $T = 0.5$ (6 months)
- $\sigma = 0.20$

Black-Scholes Formula for Call Option:

$$C = S \cdot N(d_1) - K e^{-rT} \cdot N(d_2)$$

$$d_1 = \frac{\ln(S/K) + (r + \frac{\sigma^2}{2})T}{\sigma \sqrt{T}} = \frac{\ln(100/105) + (0.05 + 0.5 \cdot 0.2^2) \cdot 0.5}{0.2 \cdot \sqrt{0.5}} \approx -0.07$$

$$d_2 = d_1 - \sigma \sqrt{T} = -0.07 - 0.2 \cdot \sqrt{0.5} \approx -0.21$$

From given:

- $N(d_1) = 0.5379$
- $N(d_2) = 0.4990$

Now calculate:

$$C = 100 \cdot 0.5379 - 105 \cdot e^{-0.05 \cdot 0.5} \cdot 0.4990 = 53.79 - 105 \cdot 0.9753 \cdot 0.4990 = 53.79 - 51.027 \approx 2.76$$
